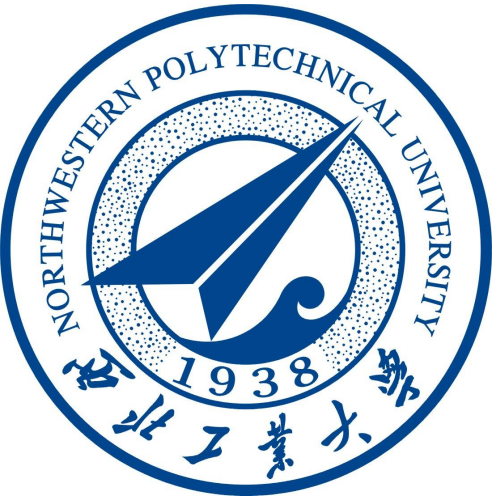


XGait: A Multi-Modality Wireless Sensing Dataset for Indoor Human Tracking and Identification

Wei Xu, Zhu Wang, Yifan Guo, Changlong Cheng, Yin Zhang,
Zhihui Ren, Bin Guo, Zhiwen Yu



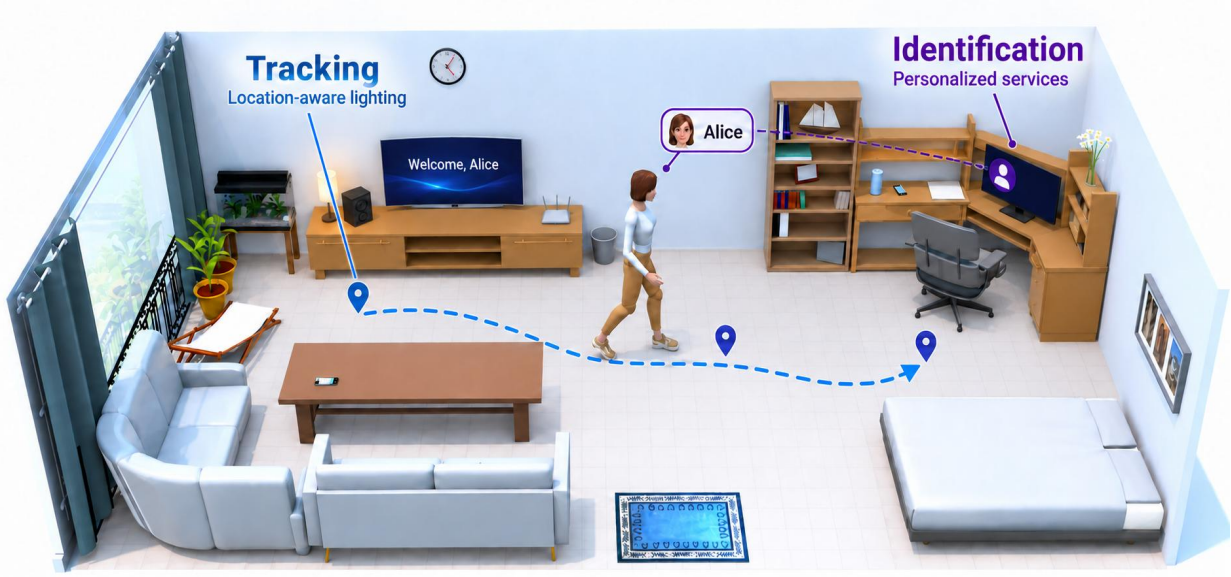
Oct. 13, 2026
Shanghai, China

UbiComp
ISWC 2026

Indoor Tracking and Identification

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- **Tracking: Where is the person?**
  - Estimate the walking trajectory.
  - Support indoor automation and movement monitoring
- **Identification: Who is walking?**
  - Recognize a person from gait.
  - Support services for the right user.



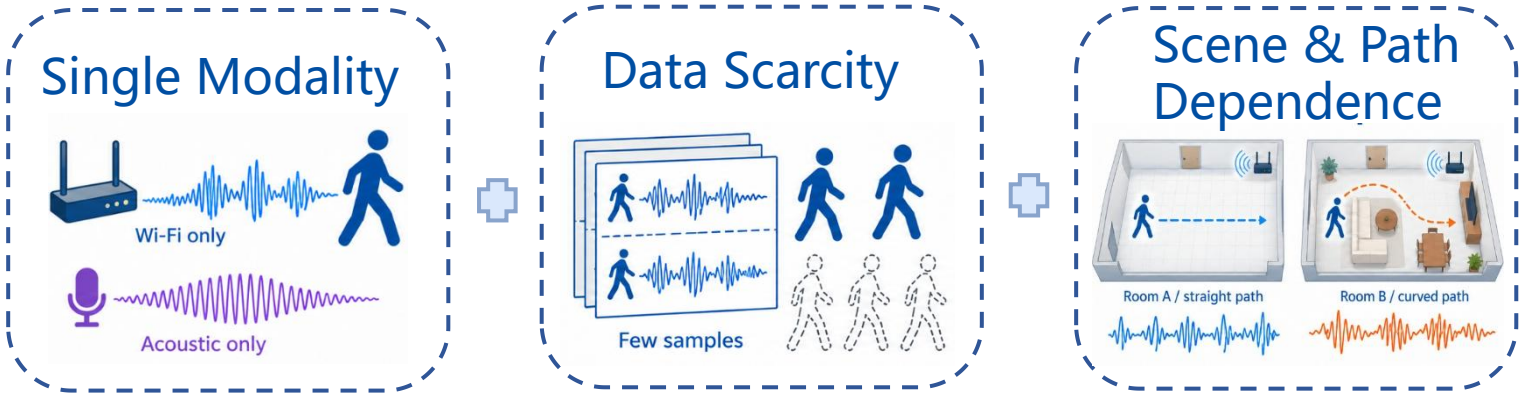
Potential setting: a smart home

Location and identity together can support personalized indoor services.

Wireless Gait Sensing

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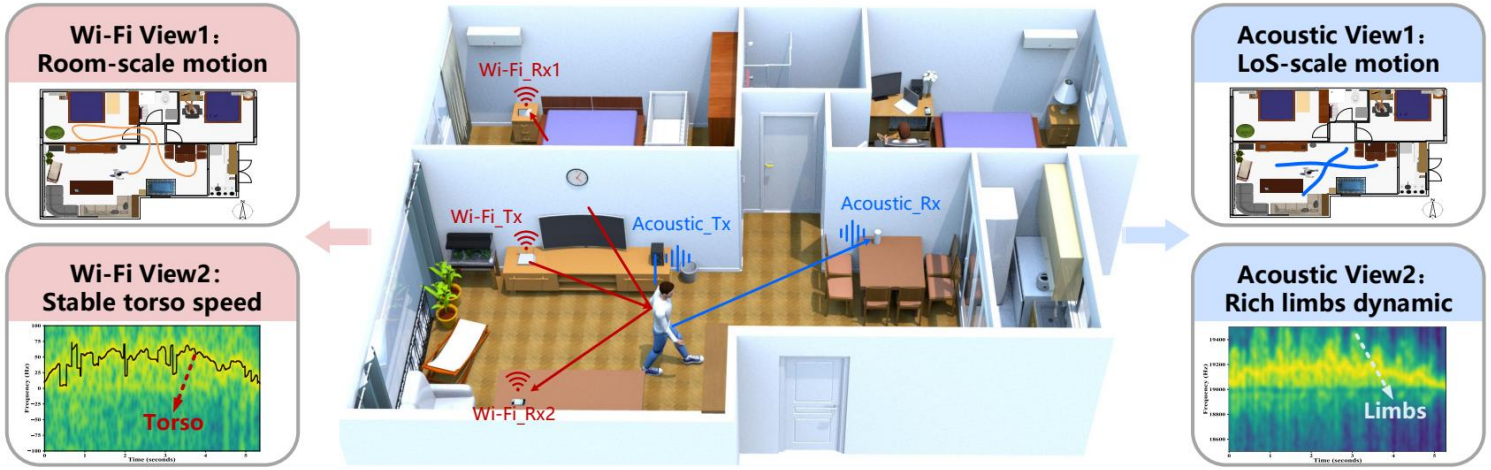
Challenges: From Laboratory to Real-world Deployment



Existing studies mostly target **constrained paths and scenarios**, leaving a significant gap from real-world deployment.

Can multiple modalities help?

- Wireless signals are fragile.
- Different signals have advantages and disadvantages.
- Can they complement each other's strengths?

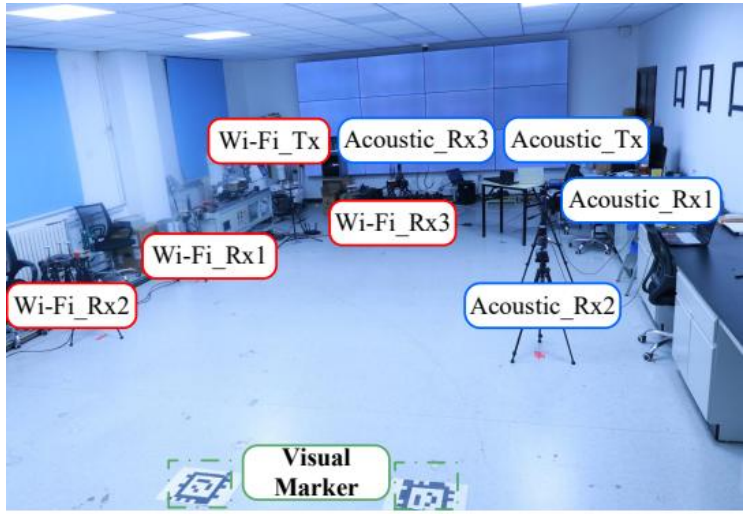


XGait Dataset Overview

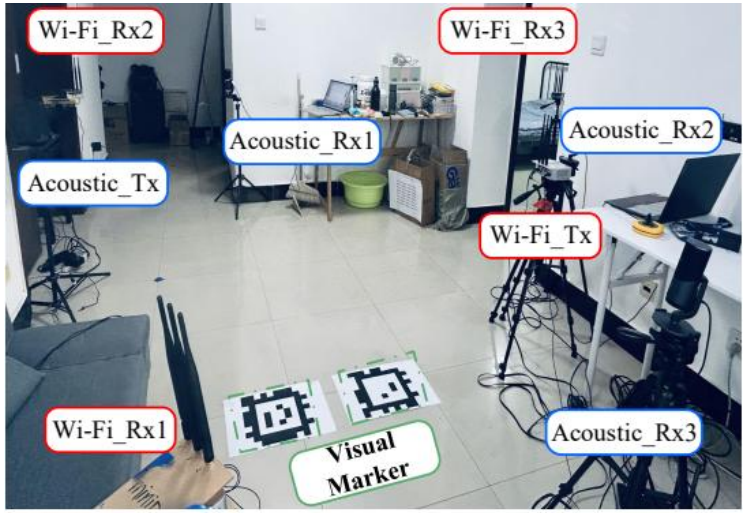
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- **27 volunteers**
- **3 indoor settings**
- **22K+ samples**

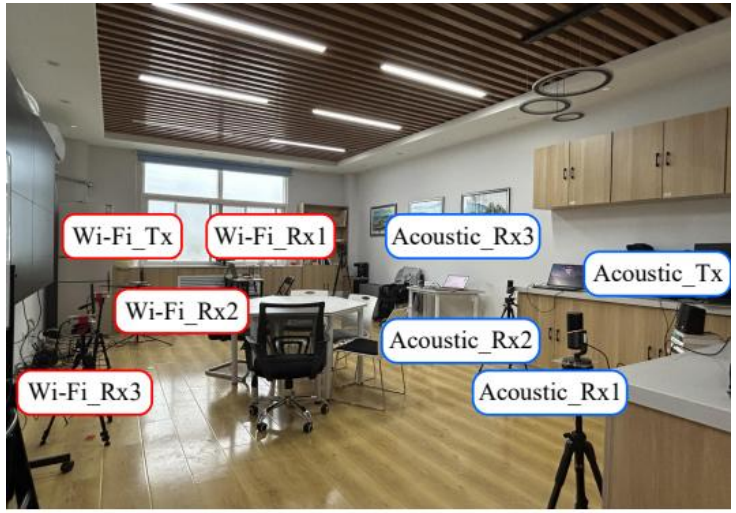
22K+ valid samples per modality. Participant groups overlap across rooms.



Laboratory



Home



Meeting room

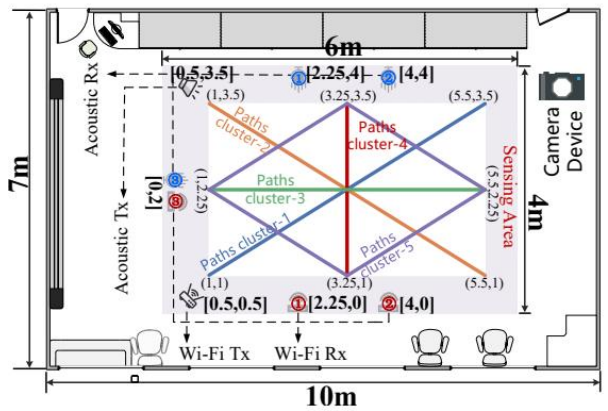
Paired wireless measurements support both tracking and identification.

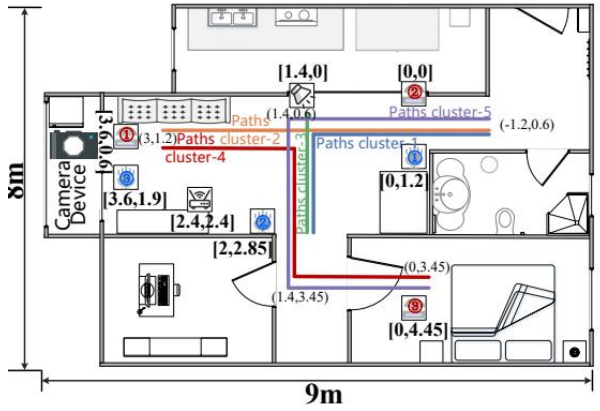
Collection and Ground Truth

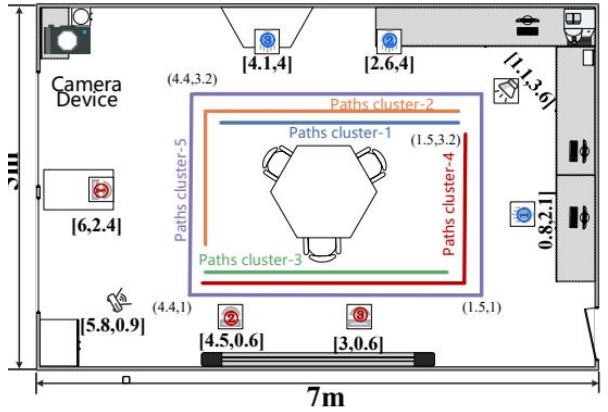
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Walking path design

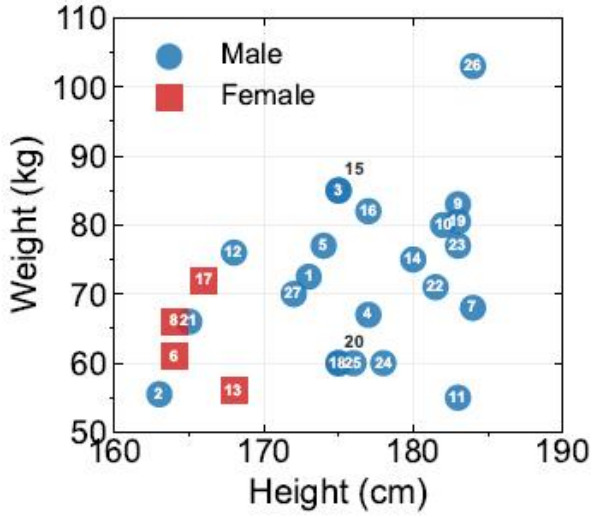
Video supplies reliable trajectory labels where visual coverage permits.

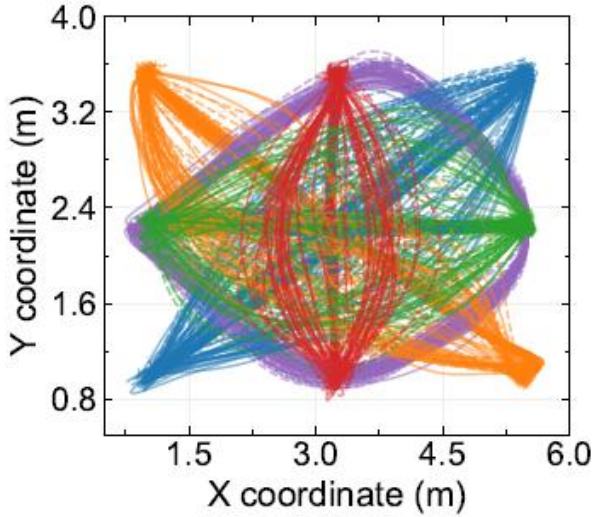


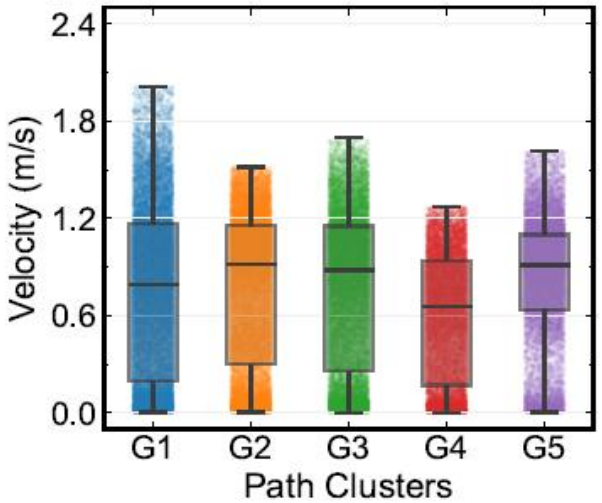




Volunteer attributes and motion diversity







Collection and Ground Truth

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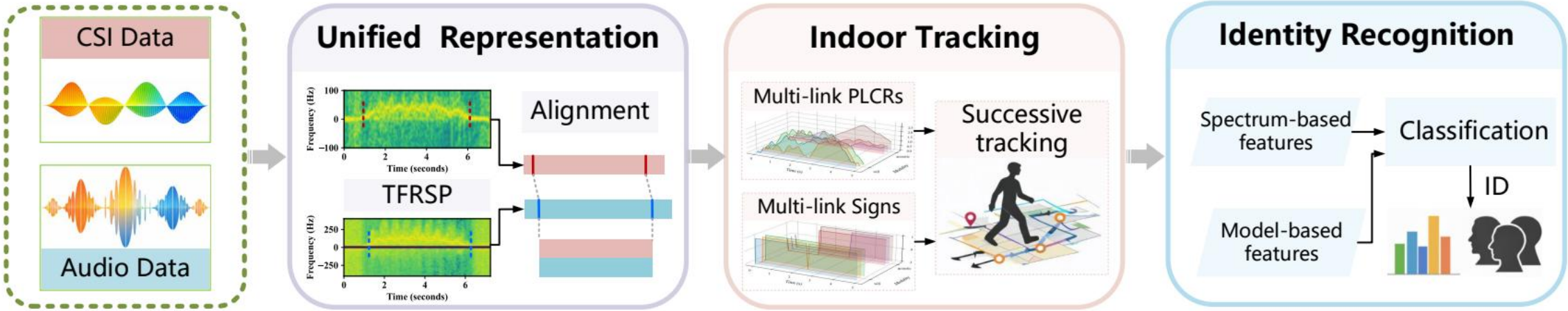
Acquisition

Wi-Fi	Active acoustics	Vision ground truth	Coarse Synchronization
1 Tx + **3** Rx	**1** Tx + **3** Rx	**1080**p, **30** fps	1.Software trigger 2.Residual timing shifts 3. torso-motion alignment
5.31 GHz	**19** kHz CW	AprilTag calibration	
1 kHz	**48** kHz	foot position –> torso position	

Dataset statistics

Scenario (subjects)	Wi-Fi	Acoustics	Vision / GT
Laboratory (**20**)	10,816	10,823	10,730
Home (**19**)	4,753	4,750	1,848
Meeting room (**20**)	5,005	4,835	2,000
Multi-day subset (**6**)	1,710	1,710	679
Total	**22,284**	**22,118**	**15,257**

A Shared Benchmark Pipeline



☐ Singal processing

- ➢ Remove noise.
- ➢ Generate spectra.
- ➢ Align motion.

☐ Tracking

- ➢ Extract multi-link PLCRs.
- ➢ Extract multi-link signs.
- ➢ Trajectory estimation.

☐ Identification

- ➢ Spectrum features (Doppler).
- ➢ Model-based features (PPVP).
- ➢ Concat fusion learning.

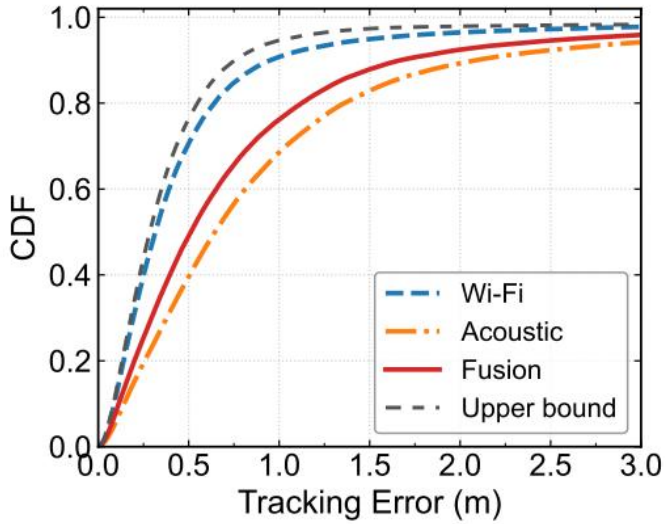
A common pipeline enables Wi-Fi, acoustic, and fusion comparisons.

Tracking Performance

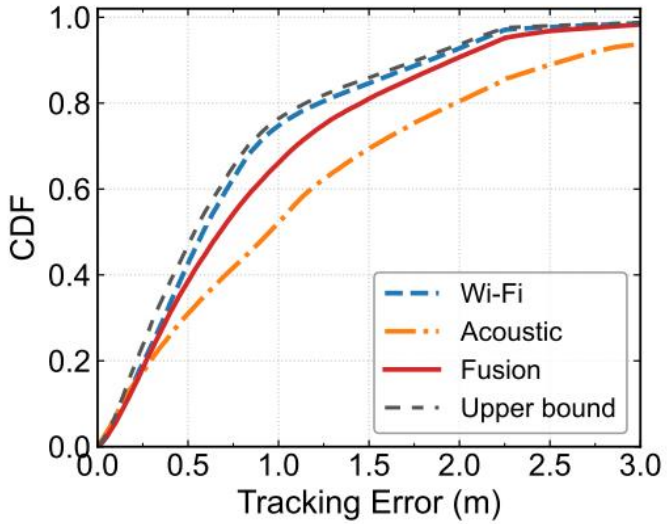
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Overall evaluation

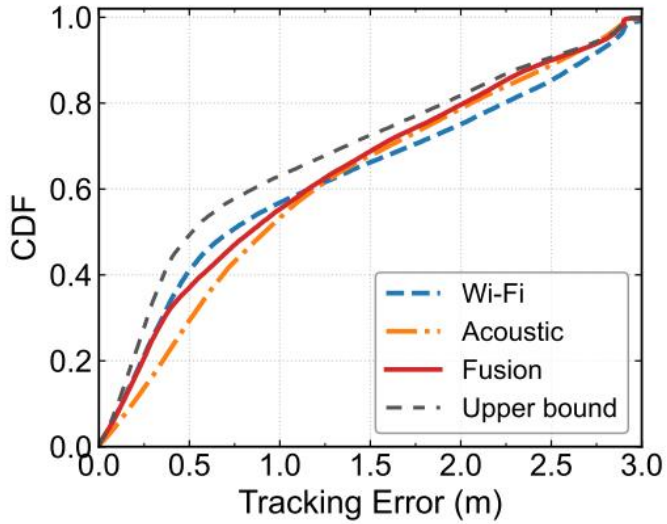
Metric: distance between an estimated position and the ground truth.



Laboratory



Home



Meeting-room

Laboratory: about 90% of Wi-Fi positions fall within 1 m.

Acoustics reaches about 70%. Evaluation uses four shared participants.

Wi-Fi gives the stronger baseline.

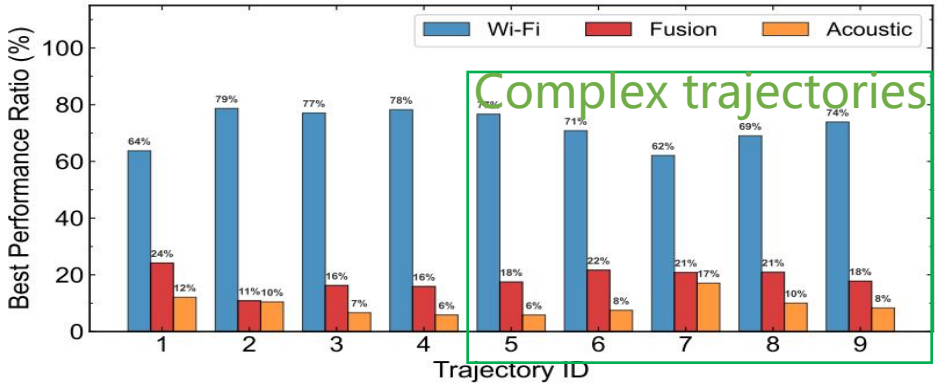
Tracking Performance

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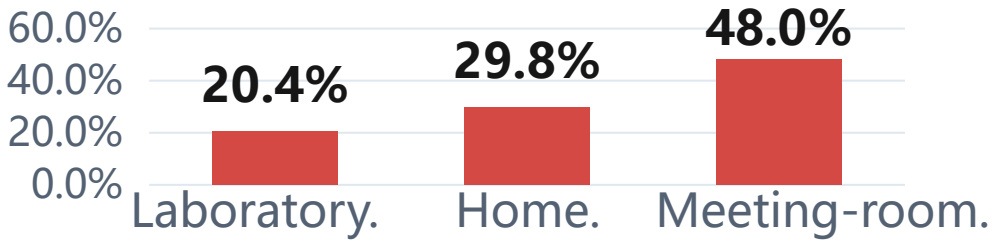
When does fusion help tracking?

Metric: trials where fusion has the lowest mean trajectory error.

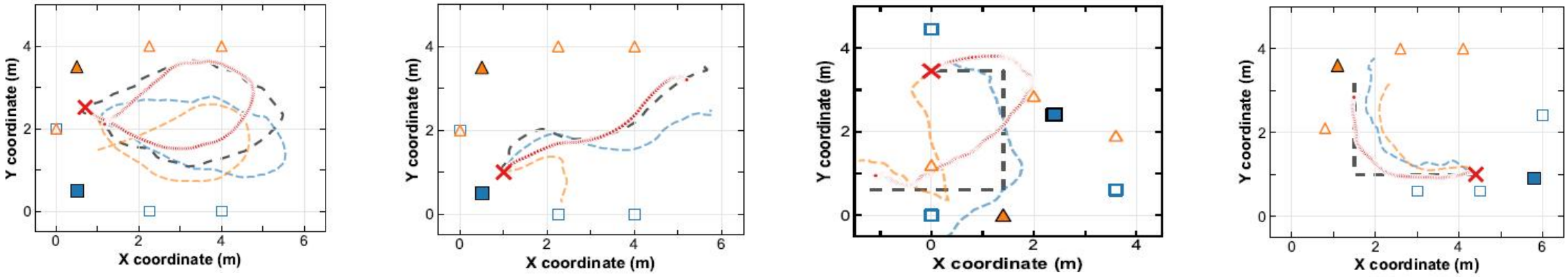
- Fusion helps on a subset of walking trails.
- The benefit varies with the sensing settings.



Different trajectory in Laboratory.



Example trajectories



Reliable fusion needs to recognize when each modality adds useful evidence.

Identification Performance

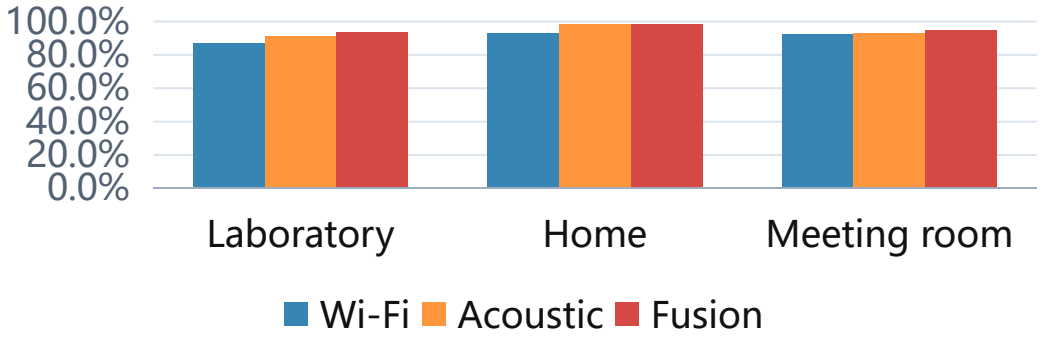
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Overall evaluation

Random split 6:2:2. Known identities: 20 in Lab, 19 in Home, 20 in Meeting room.

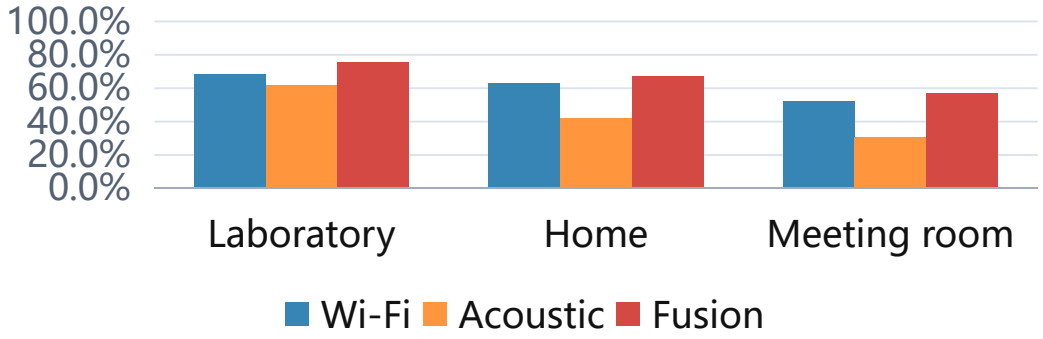
Mean accuracy across five backbones, computed within each scenario.

Doppler features



Mean (%)	Lab	Home	Meeting
Wi-Fi	87.2	93.3	92.5
Acoustic	91.1	98.1	93.3
Fusion	**93.9**	**98.2**	**94.9**

PPVP features



Mean (%)	Lab	Home	Meeting
Wi-Fi	68.4	63.1	52.3
Acoustic	62.1	42.4	31.0
Fusion	**75.7**	**67.4**	**57.0**

Fusion is strong overall. Doppler features give higher accuracy than PPVP.

Identification Performance

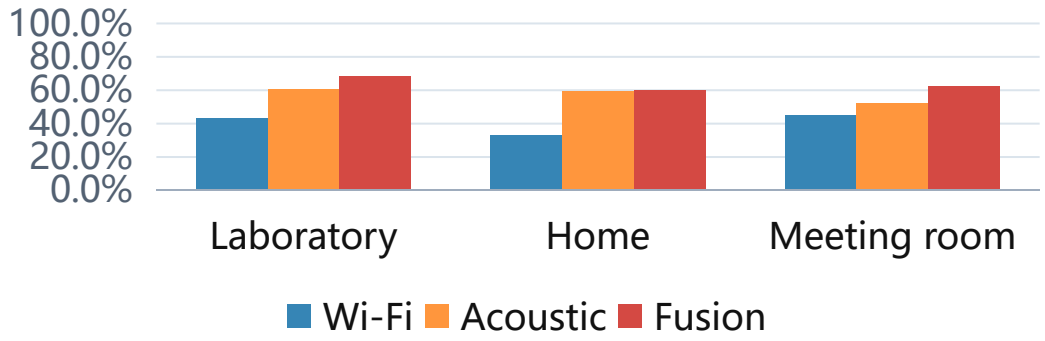
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Generalization to unseen paths

Test walks use paths excluded from training. The identities are still known.

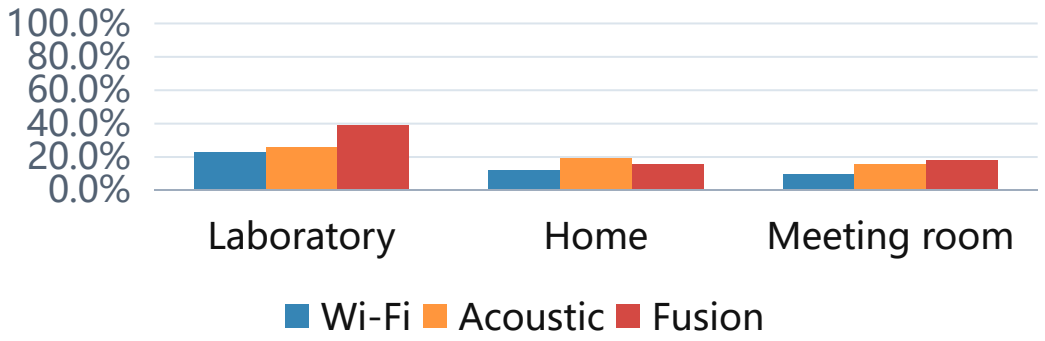
Same five backbones and feature families. Values show the mean across five method.

Doppler features



Mean (%)	Lab	Home	Meeting
Wi-Fi	43.1	33.3	45.1
Acoustic	60.8	59.6	52.3
Fusion	**68.2**	**60.2**	**62.4**

PPVP features



Mean (%)	Lab	Home	Meeting
Wi-Fi	22.8	12.1	9.8
Acoustic	25.9	19.3	16.0
Fusion	**39.3**	**16.0**	**18.3**

New paths expose a large gap. Fusion does not always beat the best modality.

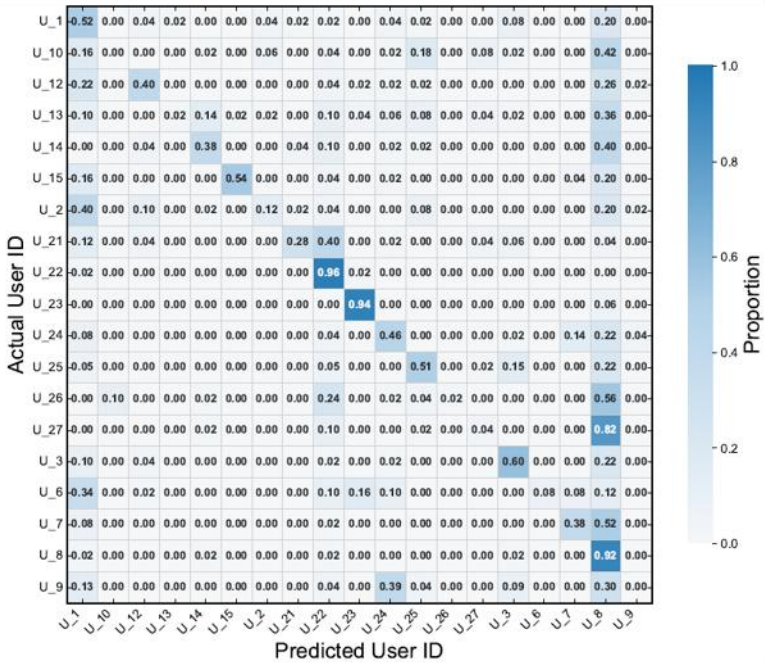
Identification Performance

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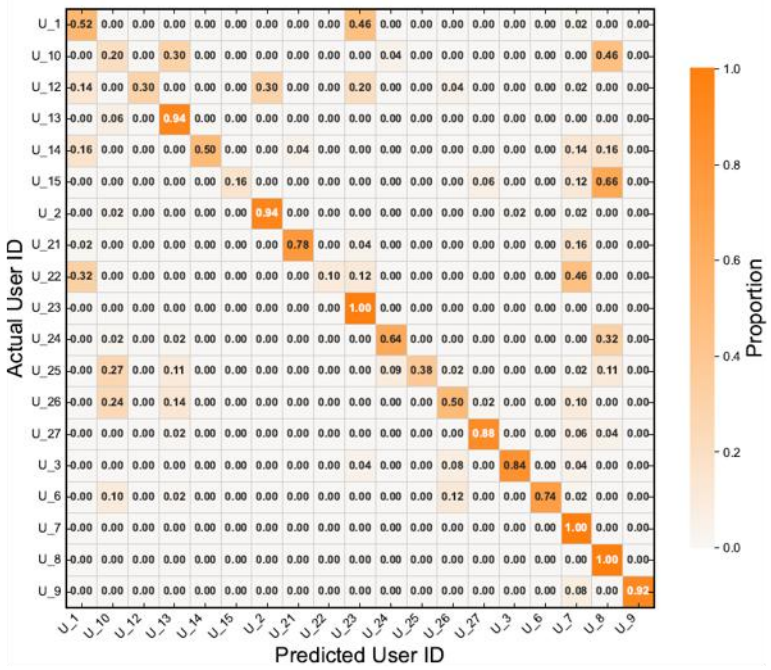
☐ Confusion matrices on unseen paths

Home, 19 known identities, CNN with Doppler-based features.

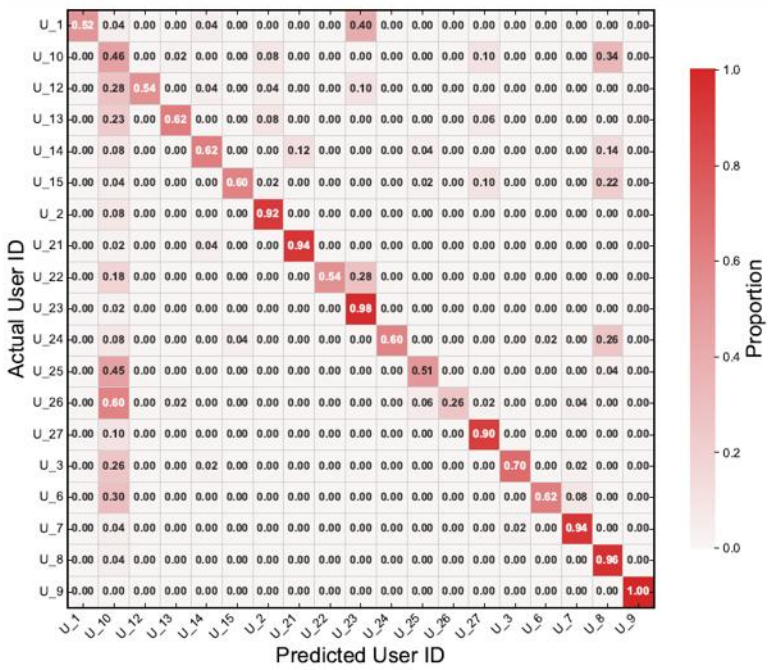
- **Wi-Fi 38.9%**



- **Acoustic 64.0%**



- **Fusion 68.7%**



Fusion gains 4.7% over acoustics.

Several single-modality errors are corrected, while some confusion remain.

Identification Performance

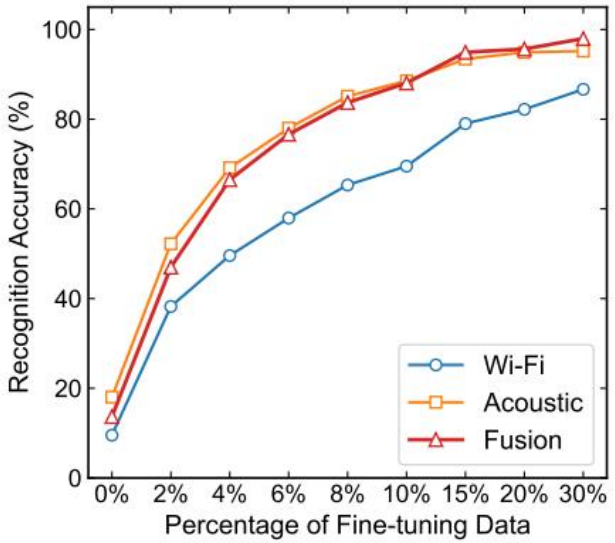
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New rooms and different days

Cross-scene evaluation uses 10 shared identities and labeled target-room data.

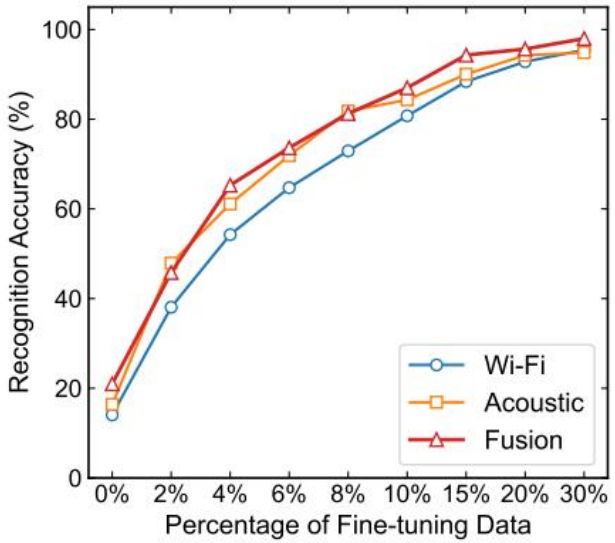
Different days evaluation users 6 shared identities in Meeting-room.

Lab. to Home



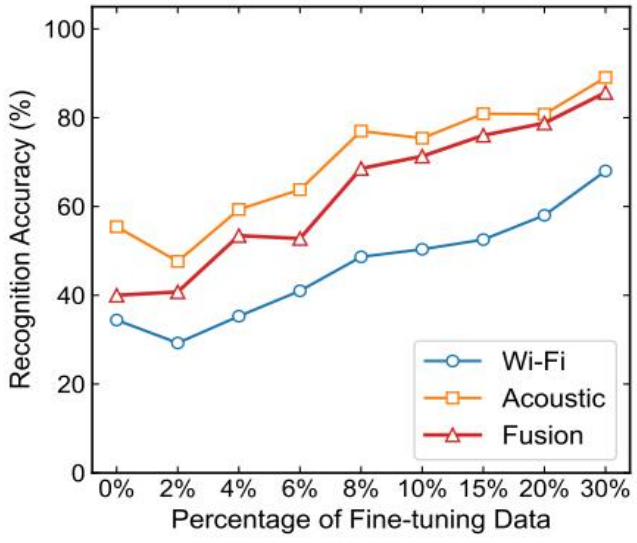
No target-room training: near chance.
A little labeled target data helps.

Lab. to Meeting room



Even 2% target-room labels help.
Fusion stays close to the best modality.

Different days



Acoustics can outperform fusion
across different days and clothing.

Multi-modal sensing still needs reliable fusion and adaptation to change.

Findings

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What do the experiments reveal?

Finding	Evidence and interpretation
1. Task-dependent strengths	Wi-Fi provides the stronger tracking baseline. Acoustics preserves more identity cues on unseen paths.
2. Conditional fusion gains	Tracking win rate varies from 20.4% to 48.0% across settings. Acoustics beats fusion in Home PPVP on unseen paths.
3. Feature choice matters	Doppler features outperform PPVP in the reported tests. PPVP carries errors from trajectory estimation.
4. Protocol-dependent accuracy	Random-split fusion Doppler accuracy averages 93.9–98.2%. Transfer to a new room starts near the 10% chance level.

Cross-scene changes include both room geometry and node placement.

Reliable sensing needs to account for the task, geometry, and signal quality.

Opportunities

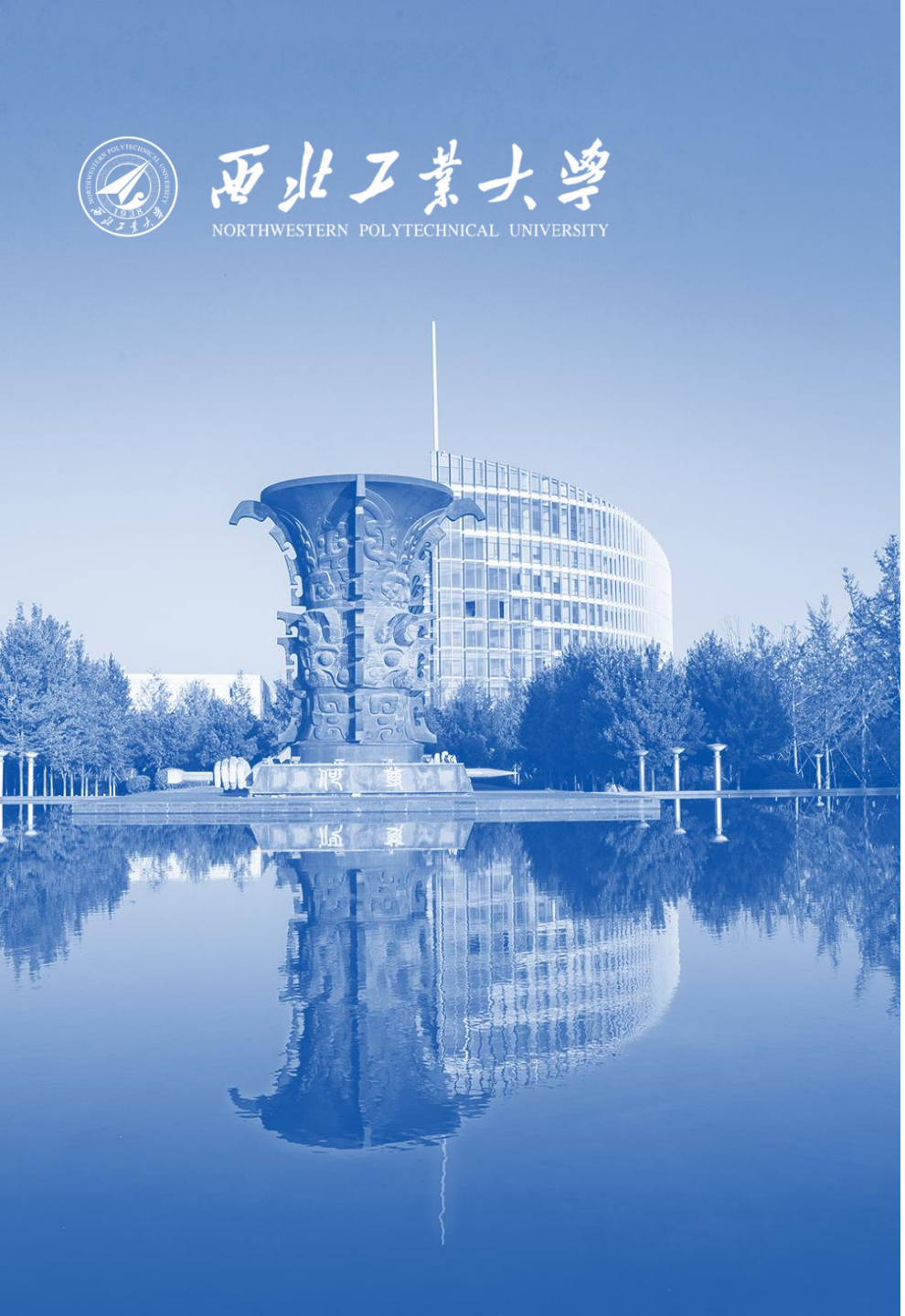
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Research questions enabled by XGait

Opportunity	Research direction and question
1. Joint motion representations	Use trajectory and heading to stabilize gait features. How can tracking and identification improve each other?
2. Reliability-aware fusion	Weight modalities and links as signal quality changes. When should a model trust or ignore each channel?
3. Adaptation with fewer labels	Study cross-modal pretraining and few-shot calibration. Can a new room work with much less labeled data?
4. Broader, lighter deployment	Evaluate more people, daily activities, and irregular layouts. Can compact models work reliably on IoT devices?

Current scope: single-person walking, 27 young adults, and stationary sensors.

XGait provides paired data to study these questions in a shared benchmark.



DATASET

CODE

EMAIL

Thanks for Listening.

Q&A